Enhancing Permeability Prediction of Heterobifunctional Degraders Using Ensemble-Derived 3D Molecular Descriptors

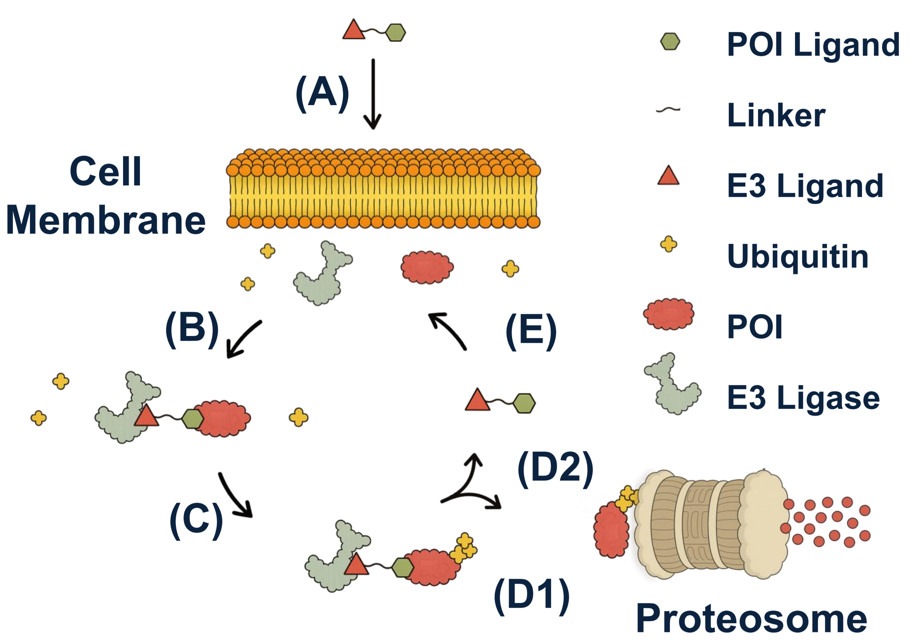
Bryant Kim[1], Robert P. Sheridan[1], Ruibo Zhang[1], Emilia Pecora de Barros[2], Jennifer Johnston[1], Li Xiao[1]

[1] Merck & Co., Inc., Rahway, NJ, United States

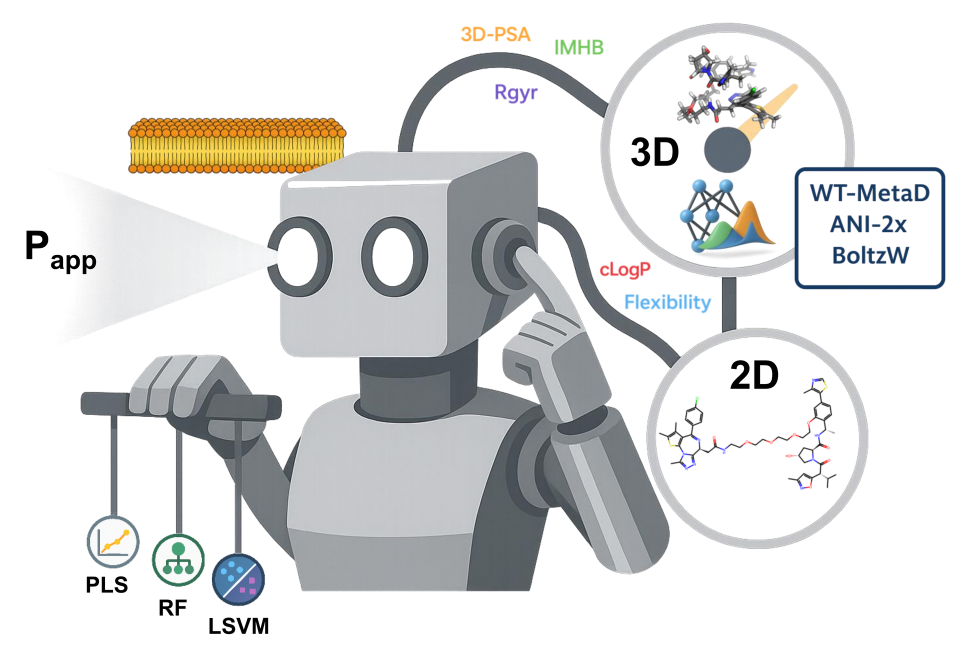
[2] MSD, London, United Kingdom

Background

Heterobifunctional degraders occupy beyond-rule-of-five chemical space, where large size, high flexibility, and environment-dependent polarity limit the applicability of traditional permeability models.[1-3] Many existing approaches rely on two-dimensional (2D) descriptors that do not capture conformational folding, polarity shielding, or three-dimensional (3D) shape. Because degraders can adopt folded conformations that effectively shield polar surface area, permeability-relevant behavior is better described by conformational ensembles rather than single static structures (**Scheme 1**). This study integrates enhanced molecular dynamics sampling in a membrane-mimetic environment with machine-learning (ML) models to relate ensemble-derived 3D molecular properties to experimentally measured permeability, as summarized in the **modeling framework**.[6]



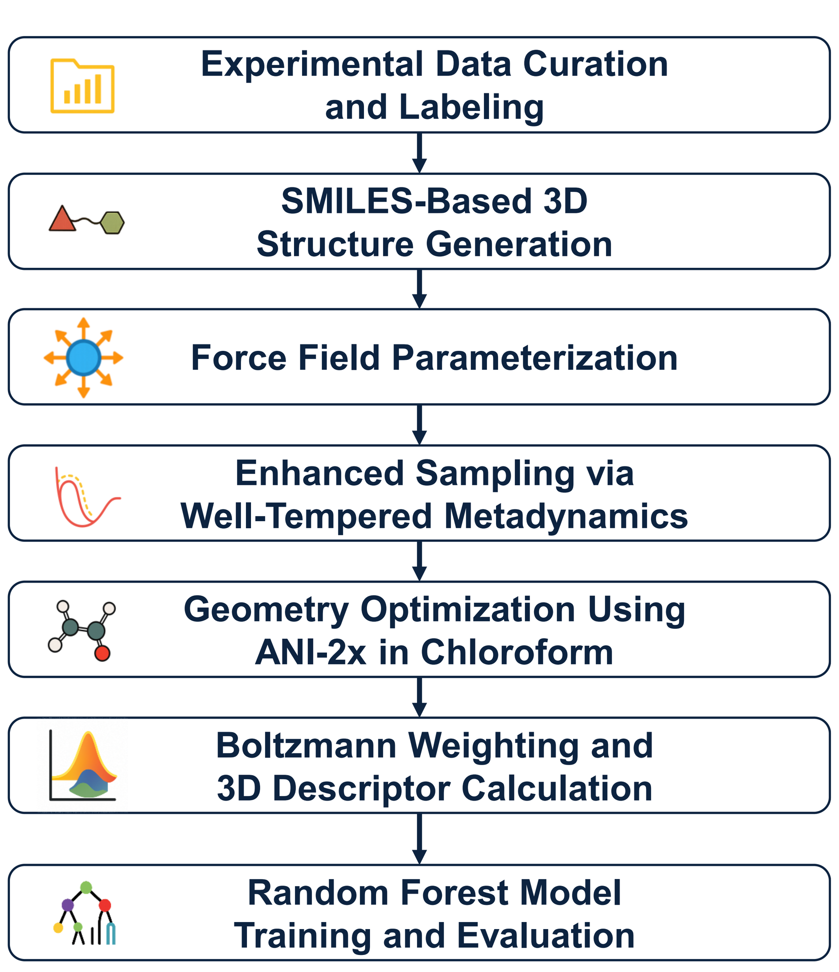
Scheme 1. Mechanism of targeted protein degradation by a heterobifunctional degrader.



Modeling Framework. Assessing ML-models for permeability using 2D and ensemble-derived 3D descriptors.

Workflow

Conformational ensembles were generated using well-tempered metadynamics (WT-MetaD) in explicit chloroform to approximate a low-dielectric membrane environment (**Scheme 2**).[4] Radius of gyration (Rgyr) was used as the collective variable to sample compact and extended states. Extracted conformers were geometry-optimized using a solvent-adapted ANI-2x neural network potential and Boltzmann-weighted.[5] Ensemble-averaged 3D descriptors, including Rgyr, intramolecular hydrogen bonds (IMHB), and three-dimensional polar surface area (3D-PSA), were combined with established 2D descriptors (**Figure 1**) to train regression models for passive permeability prediction.



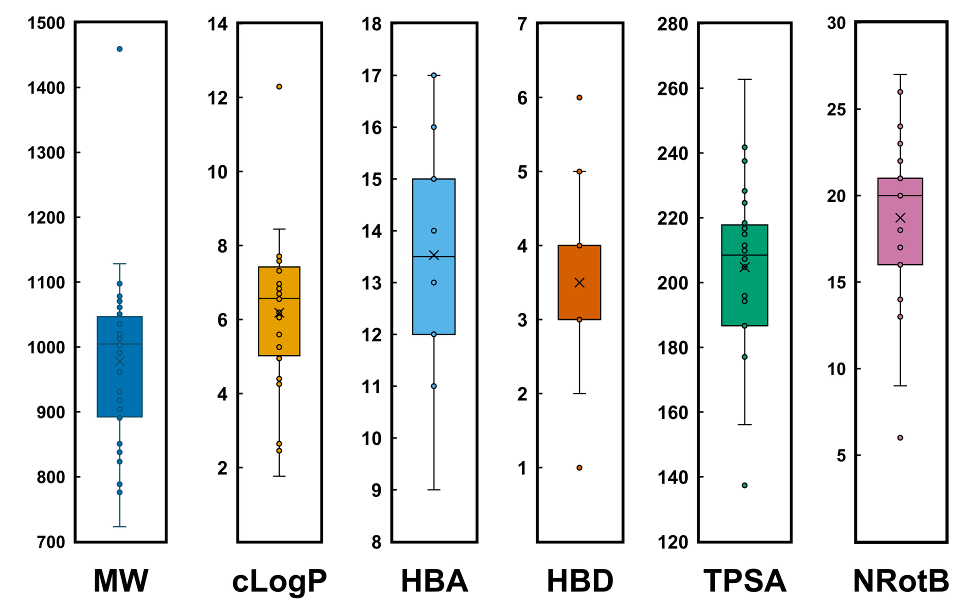


Figure 1. Distribution of calculated 2D molecular properties for the degrader data set, including size, lipophilicity, polarity, and flexibility.

Scheme 2. Computational workflow used for model development, including data curation, ensemble generation, Boltzmann-weighted 3D descriptor calculation, and ML training and evaluation.

Results

Across all evaluated ML models, inclusion of 3D descriptors improved predictive performance relative to 2D descriptors alone (**Figure 2**). Partial least-squares (PLS) models using only 3D descriptors achieved the highest performance, with a mean r^2 of ~0.48 across 100 randomized 50/50 train–test splits. Random forest (RF) and linear support vector machine (LSVM) models showed similar trends with lower overall accuracy. Models incorporating 3D descriptors consistently outperformed 2D-only models, with reduced RMSE and improved correlation. The underlying ensemble-derived 3D descriptor distributions are shown in **Figure 3**. Performance metrics are averaged over 100 randomized splits to ensure robustness under data-limited conditions.

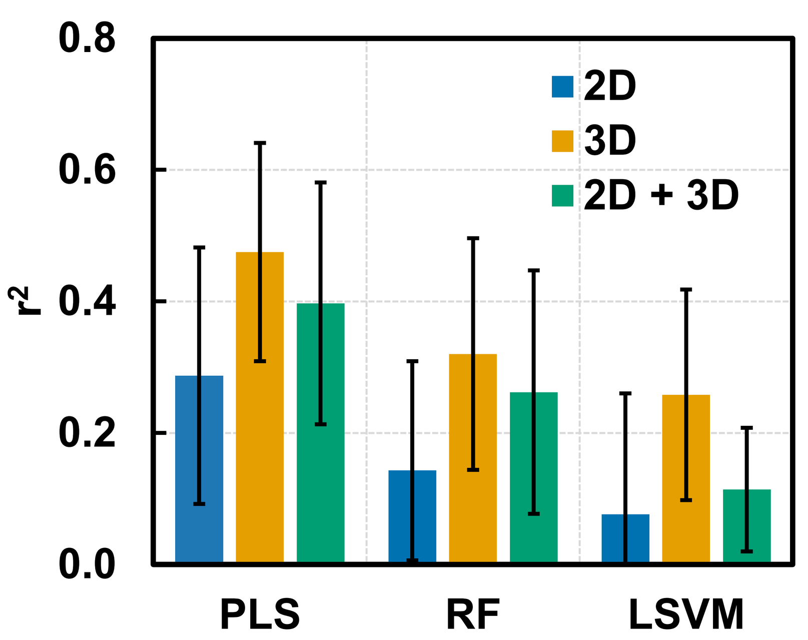


Figure 2. Predictive performance of PLS, RF, and LSVM models using 2D, 3D, and combined sets.

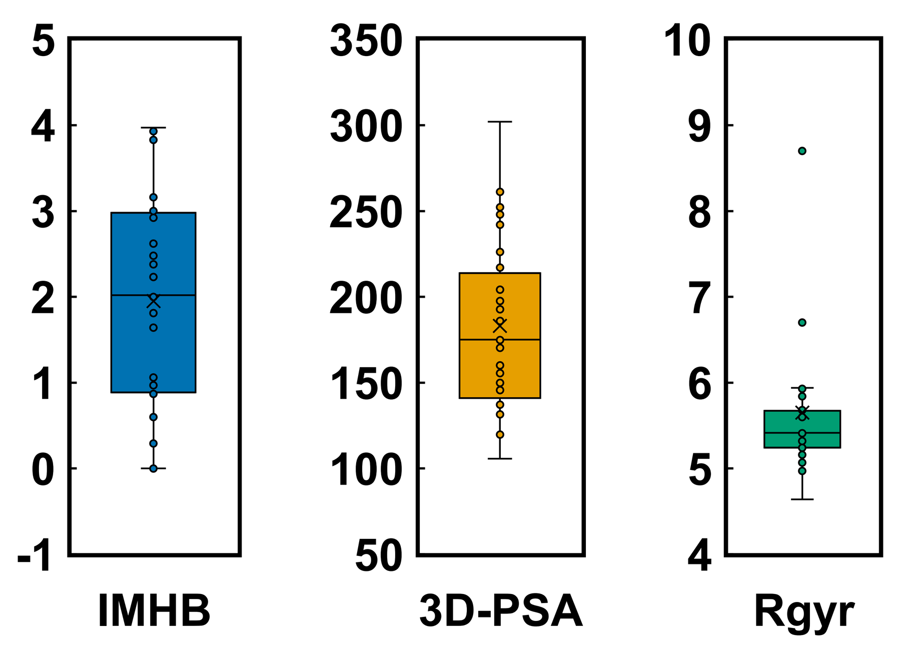


Figure 3. Boltzmann-weighted distributions of ensemble-derived 3D molecular descriptors.

Table 1. Model Performance Across Descriptor Sets

Model	Descriptors	r^2	RMSE
PLS	2D	0.29	0.61
PLS	3D	0.48	0.53
PLS	2D + 3D	0.40	0.55
RF	2D	0.14	0.64
RF	3D	0.32	0.55
RF	2D + 3D	0.26	0.58
LSVM	2D	0.08	0.63
LSVM	3D	0.26	0.58
LSVM	2D + 3D	0.11	0.61

Mean values over 100 randomized 50/50 train–test splits.

Features

Feature importance analysis identified Rgyr as the dominant predictor of permeability across all models and descriptor sets, with additional contributions from 3D-PSA and IMHB (**Figure 4**). Bivariate analysis revealed a strong linear relationship between permeability and molecular compactness, with Rgyr explaining ~50% of the variance across the data set. In contrast, 3D-PSA showed weak global correlation but stronger relationships within a polarity range.

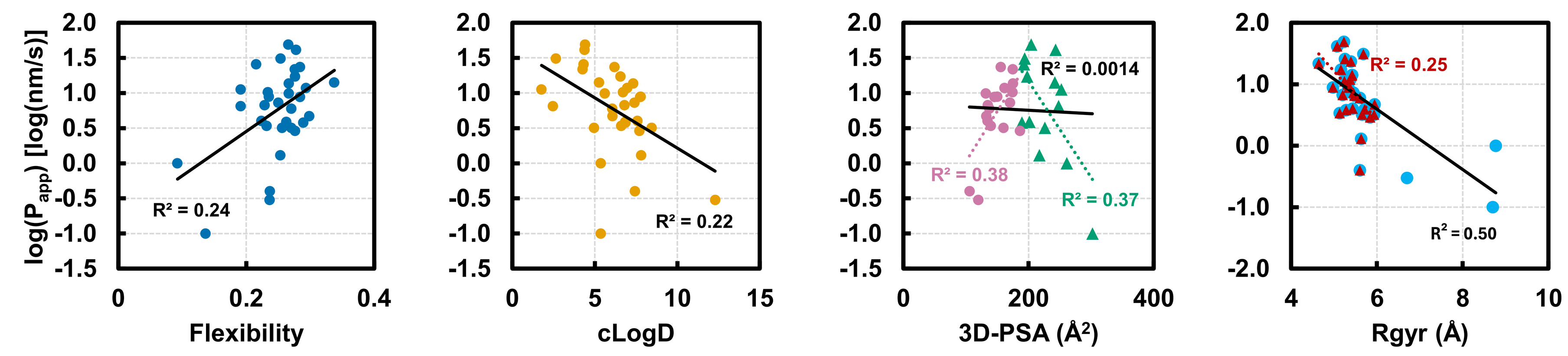


Figure 4. Relationships between key molecular descriptors and log-transformed passive permeability.

Representative low-energy conformations illustrate how Rgyr, 3D-PSA, and IMHB jointly influence permeability: highly permeable compounds adopt folded conformations with moderate polarity, whereas low-permeability compounds are either extended and highly polar or compact but insufficiently polar (**Figure 5**).

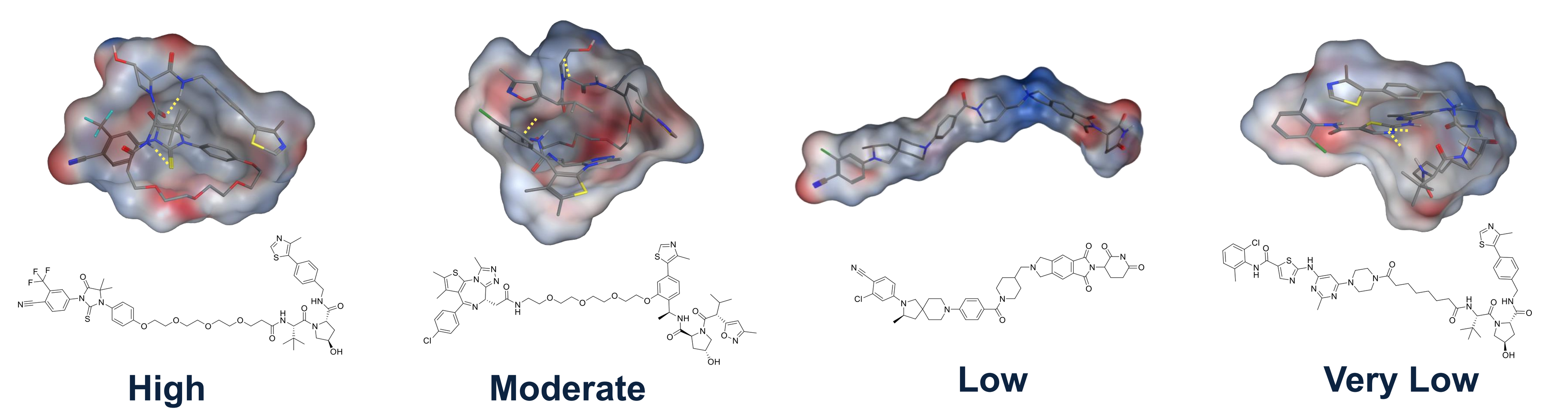


Figure 5. Representative low-energy conformations highlighting features associated with permeability.

Design

- Higher permeability is associated with compact conformations (Rgyr < ~6.5 Å).
- Optimal permeability occurs within a moderate polarity range (3D-PSA ~140–230 Å²).
- IMHBs enable polarity shielding that supports permeability.

$$R_{gyr} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(\mathbf{r}_i - \mathbf{r}_{COM})^2}$$

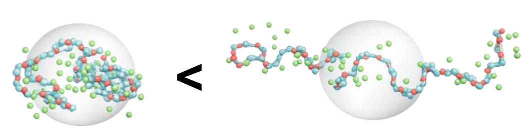


Table 2. 3D Descriptor Criteria Defining Permeability Profiles

Permeability Class	Rgyr (Å)	3D-PSA (Å²)	IMHB	Structural Description
High	4.5–5.5	170–220	1–4	Compact, polarity-shielded
Moderate	5.0–6.0	135–190	1–3	Intermediate
Low	>6.5	>230	<1	Extended, polar
Very Low	<6.5	<140	<2	Compact, low polarity

Conclusions

Ensemble-derived 3D molecular descriptors improve passive permeability prediction for heterobifunctional degraders in beyond-rule-of-five chemical space. Models incorporating 3D descriptors consistently outperform 2D-only approaches across all evaluated machine-learning methods, with the strongest performance observed for partial least-squares regression. Molecular compactness, quantified by radius of gyration, emerges as the dominant and most interpretable predictor of permeability. The WT-MetaD + ANI-2x workflow provides physically meaningful, QSAR-relevant 3D descriptors that enable quantitative analysis of structure–permeability relationships and is broadly applicable to other conformationally complex beyond-rule-of-five modalities.[6]

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Contact information

Bryant Kim
Email: bryant.kim@merck.com
ORCID: 0000-0003-2128-7066

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